



AI-Based Smart Cardiopulmonary Blood Flow Control System

Fuzzy Logic Controller Report

Student Name: Sawab Hussein

Instructor: Dr. Omar Abdulghafoor

Course: Artificial Intelligence, Spring 2026

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1 Problem Statement

A cardiopulmonary bypass rotary pump must maintain safe blood flow by adjusting the pump speed N . The required operating point is

$$\Delta P = 84 \text{ mmHg}, \quad Q = 6 \text{ L/min.}$$

The work completed for this project was to convert the assignment rule table into a complete fuzzy controller, evaluate the controller at the required operating point, draw the intermediate and final fuzzy sets, and compute the crisp rotary-pump speed. The implementation in `fuzzy_cpb_controller.py` defines the membership functions, stores the nine IF–THEN rules, performs Mamdani inference, and applies centroid defuzzification by numerical integration.

The controller uses the standard Mamdani sequence: fuzzify the two inputs, compute each rule firing strength with the logical AND operator as a minimum, clip each output membership function by its rule strength, combine all output clips with a maximum operator, and finally defuzzify the aggregated output set.

2 Membership Functions and Rules

The triangular membership functions were read from the assignment figures. The universes are $Q \in [0, 10]$ L/min, $\Delta P \in [50, 150]$ mmHg, and $N \in [2800, 3200]$ rpm; the vertices used in the implementation are summarized below.

Table 1: Triangular membership vertices used by the controller.

Variable	Low / Slow	Medium	High / Fast
Blood flow (L/min)	$Q (0, 3.5, 6)$	$(3.5, 6, 8.5)$	$(6, 8.5, 10)$
Delta pressure (mmHg)	$\Delta P (50, 72, 100)$	$(72, 100, 120)$	$(100, 120, 142)$
Pump speed (rpm)	$N (2800, 2900, 3000)$	$(2900, 3000, 3100)$	$(3000, 3100, 3200)$

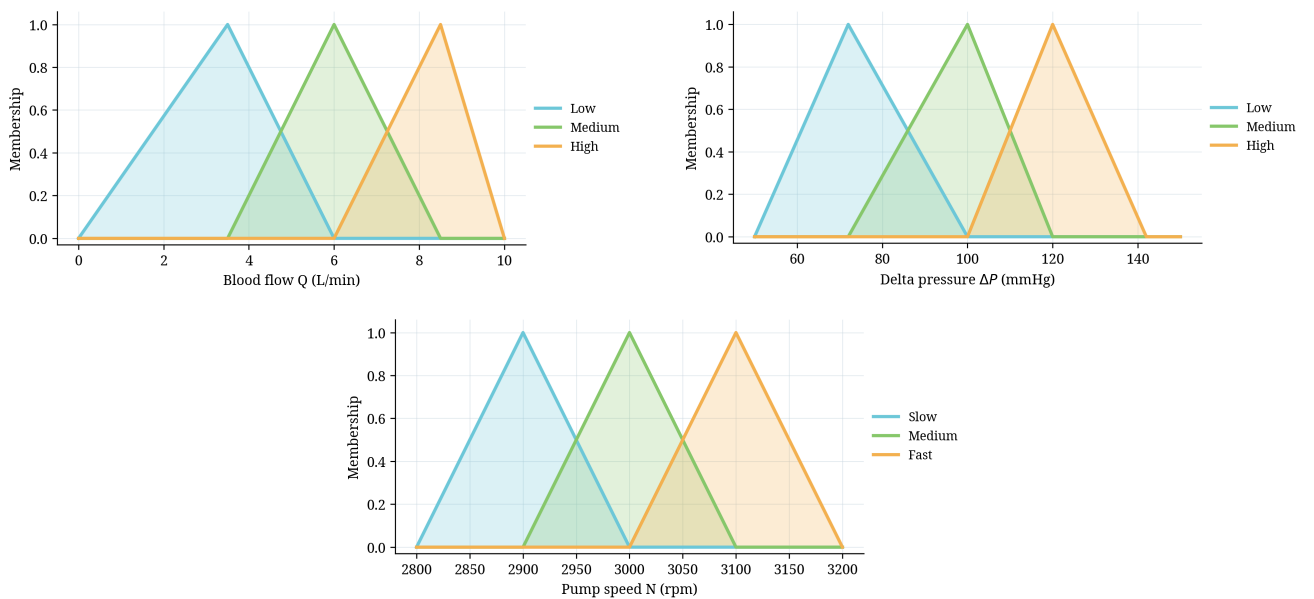


Figure 1: Membership functions for flow, pressure difference, and pump speed.

Figure 1 is the controller definition in graphical form. The first two panels show how measured flow and pressure are translated into linguistic values, while the third panel shows the possible

linguistic pump-speed outputs. These curves are the numerical basis used by the Python file when it calculates memberships for $Q = 6$ and $\Delta P = 84$.

Table 2: Fuzzy rule base used for the rotary pump controller.

$\Delta P \backslash Q$	Low Flow	Medium Flow	High Flow
Low ΔP	Fast	Medium	Slow
Medium ΔP	Medium	Medium	Slow
High ΔP	Medium	Medium	Slow

Explicit IF–THEN Rules

1. IF ΔP is low AND Q is low THEN Speed = fast
2. IF ΔP is low AND Q is medium THEN Speed = medium
3. IF ΔP is low AND Q is high THEN Speed = slow
4. IF ΔP is medium AND Q is low THEN Speed = medium
5. IF ΔP is medium AND Q is medium THEN Speed = medium
6. IF ΔP is medium AND Q is high THEN Speed = slow
7. IF ΔP is high AND Q is low THEN Speed = medium
8. IF ΔP is high AND Q is medium THEN Speed = medium
9. IF ΔP is high AND Q is high THEN Speed = slow

3 Inference at the Required Operating Point

The input $Q = 6$ lies exactly at the peak of the medium-flow triangle. Therefore, the flow input belongs completely to the medium set and has zero membership in the low and high sets. The pressure value $\Delta P = 84$ lies in the overlap between low and medium pressure, so both pressure labels are partially active.

For $Q = 6$, the medium-flow set is fully active:

$$\mu_{Q,\text{low}}(6) = 0, \quad \mu_{Q,\text{medium}}(6) = 1, \quad \mu_{Q,\text{high}}(6) = 0.$$

For $\Delta P = 84$, the two active pressure memberships are

$$\mu_{\Delta P,\text{low}}(84) = \frac{100 - 84}{100 - 72} = 0.571, \quad \mu_{\Delta P,\text{medium}}(84) = \frac{84 - 72}{100 - 72} = 0.429.$$

Only two rules fire with nonzero strength:

Table 3: Nonzero rule activations.

Rule	Consequent	Firing strength
R2	Low ΔP , Medium Q : Speed is Medium	0.571
R5	Medium ΔP , Medium Q : Speed is Medium	0.429

Both active rules point to the same consequent, so the aggregated output is simply the medium-speed membership function clipped at

$$\alpha_{\text{medium}} = \max(0.571, 0.429) = 0.571.$$

The remaining rules have zero firing strength because one of their required input labels has zero membership at this operating point.

Figure 2 visualizes the full rule matrix and confirms that only two cells are active: low pressure with medium flow, and medium pressure with medium flow. Figure 3 then shows the consequence of these active rules: the medium-speed output set is clipped at 0.571, while the slow and fast out-

put sets do not contribute to the aggregate. The vertical dashed line marks the centroid of the aggregated output.

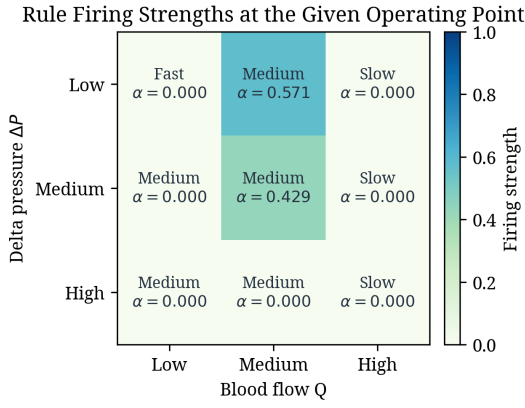


Figure 2: Rule firing strengths.

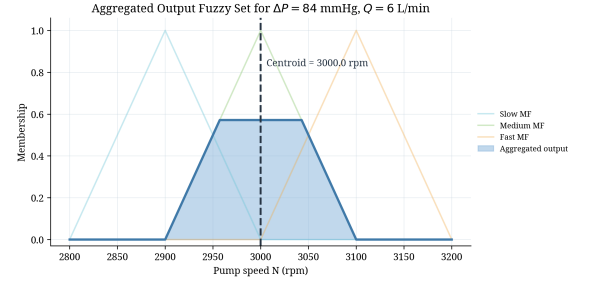


Figure 3: Aggregated output fuzzy set.

4 Centroid Defuzzification

The centroid method computes

$$N^* = \frac{\int_{2800}^{3200} n \mu_{\text{agg}}(n) dn}{\int_{2800}^{3200} \mu_{\text{agg}}(n) dn}.$$

For this operating point, the active output is the medium-speed triangle clipped at $\alpha = 0.571 = 4/7$. The unclipped medium-speed set is centered at 3000 rpm and has the vertices (2900, 0), (3000, 1), and (3100, 0). The clipping level intersects the triangle at

$$n_L = 2900 + 100 \left(\frac{4}{7} \right) = 2957.143, \quad n_R = 3100 - 100 \left(\frac{4}{7} \right) = 3042.857.$$

Therefore, the aggregated membership function is

$$\mu_{\text{agg}}(n) = \begin{cases} \frac{n - 2900}{100}, & 2900 \leq n \leq 2957.143, \\ \frac{4}{7}, & 2957.143 \leq n \leq 3042.857, \\ \frac{3100 - n}{100}, & 3042.857 \leq n \leq 3100, \\ 0, & \text{otherwise.} \end{cases}$$

First, evaluate the denominator, which is the area under the clipped output set:

$$\begin{aligned} A &= \int_{2900}^{2957.143} \frac{n - 2900}{100} dn + \int_{2957.143}^{3042.857} \frac{4}{7} dn + \int_{3042.857}^{3100} \frac{3100 - n}{100} dn \\ &= 16.327 + 48.980 + 16.327 \\ &= 81.633. \end{aligned}$$

Next, evaluate the numerator, which is the first moment of the clipped output set:

$$\begin{aligned} M &= \int_{2900}^{2957.143} n \left(\frac{n - 2900}{100} \right) dn + \int_{2957.143}^{3042.857} n \left(\frac{4}{7} \right) dn + \int_{3042.857}^{3100} n \left(\frac{3100 - n}{100} \right) dn \\ &= \left[\frac{n^3}{300} - \frac{29n^2}{2} \right]_{2900}^{2957.143} + \left[\frac{2n^2}{7} \right]_{2957.143}^{3042.857} + \left[\frac{31n^2}{2} - \frac{n^3}{300} \right]_{3042.857}^{3100} \\ &= 47,968.902 + 146,938.776 + 49,990.282 \\ &= 244,897.959. \end{aligned}$$

Finally, divide the first moment by the area:

$$N^* = \frac{M}{A} = \frac{244,897.959}{81.633} = \boxed{3000.0 \text{ rpm}}.$$

The exact arithmetic gives the same value because $M = 12,000,000/49$ and $A = 4000/49$, so the ratio is exactly 3000.

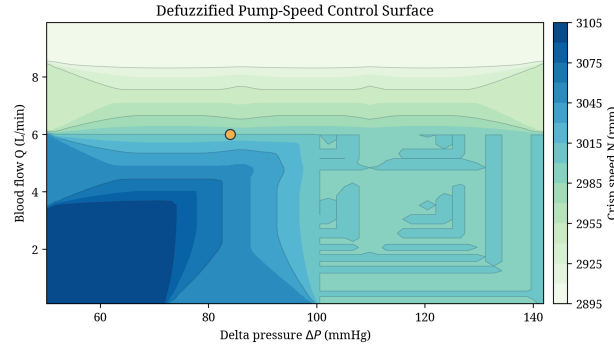


Figure 4: Defuzzified control surface with the required operating point highlighted.

Figure 4 is a broader check of the controller behavior over the input space. Each point on the surface is obtained by running the same fuzzy inference and centroid calculation for a different pair of $(\Delta P, Q)$. The highlighted point corresponds to the assignment case, showing where the computed 3000 rpm result sits relative to nearby operating conditions.

5 Conclusion

At $\Delta P = 84$ mmHg and $Q = 6$ L/min, the fuzzy controller activates the medium-speed consequent only. After aggregation and centroid defuzzification, the recommended rotary-pump speed is

$$\boxed{3000 \text{ rpm}}.$$

This value is consistent with the controller goal of maintaining the reference flow near 6 L/min under moderate pressure loading. The Python file documents the computational workflow used for the fuzzy inference and defuzzification results in this report.